

MATH – 241
Engineering Mathematics I
Problem Set – 1

Question 1. Prove that

$$|Im\{1 - 3\bar{z} + z^2\}| < 5, \text{ if } |z| < 1.$$

Note : $\bar{z}$ represents complex conjugate of z .

Question 2. Prove that

$$\sqrt{2}|z| \geq |Re(z)| + |Im(z)| \text{ for all } z \in \mathbb{C}.$$

Question 3. Prove that

$$|z|^2 + a^2 = |z + a|^2 - 2Re\{az\} \text{ for any } a \in \mathbb{R} \text{ and } z \in \mathbb{C}.$$

Question 4. Prove that

$$|z|^2 + 2Re\{bz\} = |z + \bar{b}|^2 - |b|^2 \text{ for any } b, z \in \mathbb{C}.$$

Question 5. (A) Prove that

$$|z - w| \geq ||z| - |w|| \text{ for any } w, z \in \mathbb{C}.$$

(B) If $|z| = 3$, prove that

$$\frac{1}{|z^4 - 7z^2 + 10|} \leq \frac{1}{28}$$

Question 6. Let $p(z) = a_0 + a_1z + a_2z^2 + \dots + a_nz^n$ be a polynomial where $a_0, a_1, \dots, a_n \in \mathbb{C}$ and $a_n \neq 0$. Prove that

$$|p(z)| \geq \frac{1}{2} |a_n| |z|^n$$

for $|z| \geq R$ where

$$R = \max\left\{1, 2\left(\frac{|a_0| + |a_1| + |a_2| + \dots + |a_{n-1}|}{|a_n|}\right)\right\}$$

Question 7. In defining complex numbers, we have labeled root of $x^2 + 1 = 0$ as j and generated the complex number set as $\mathbb{C} = \{z : z = \alpha + j\beta\}$ for all $\alpha, \beta \in \mathbb{R}$.

In the spirit of this, we could have used other polynomials whose roots are not real. Consider root of $x^2 + x + 1 = 0$ and label it as k . Then generate another set of number that we will call as Komplex, $\mathbf{K}$, where $y \in \mathbf{K}$ if $y = \alpha + \beta k$ for $\alpha, \beta \in \mathbb{R}$.

Define an addition and a multiplication rule so that for $\beta = 0$, the rules reduce to their ordinary summation and multiplication rules of reals, and $\mathbf{K}$ is a field.

Field properties:

$(\mathbf{K}, +, \cdot) \quad \forall u, v, w \in \mathbf{K} :$

- $u + v = v + u$
- $u + (v + w) = (u + v) + w$
- $\exists 0. \ni . 0 + u = u$
- $\exists -u . \ni . u + (-u) = 0$
- $u \cdot v = v \cdot u$
- $u \cdot (v \cdot w) = (u \cdot v) \cdot w$
- $u \cdot (v + w) = (u \cdot v) + (u \cdot w)$
- $\exists 1 . \ni . 1 \cdot u = u$
- $\forall u \neq 0, \exists u^{-1} . \ni . u \cdot (u^{-1}) = 1.$